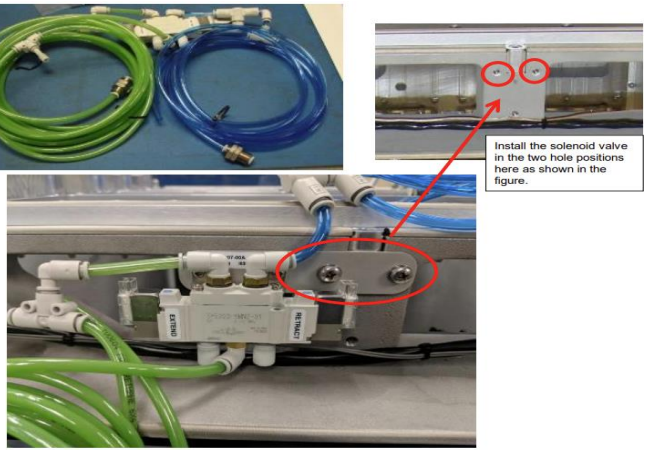
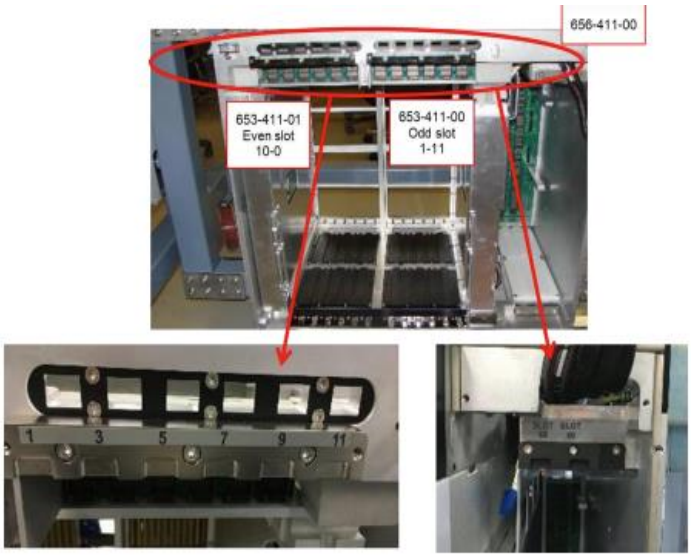
Build Overview:

1. **IPD Air Valve Install**
2. **IPD Control Cable install**
3. **IPD Bracket Install**
4. **IPD Cover Install**
5. **NVRAM Cable Install**
6. **Label Install**
7. **Optical cable Holder (bracke) install**





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	638-907-01	ELEC-MECH IPD VALVE SHUTL VER FOR SEMI TEST	1
2	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W	2
3	639-820-00	CABLE IPD CONTROL W/ CONN	1
4	813-929-01	HDW VELCRO STRP 5/8	7
5	634-338-00	ALUMINUM COVER AIR HOSE BRACKET OR MISC ARTICLE	2
6	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	6
7	813-929-00	HDW VELCRO STRP 5/8	1
8	653-411-00	PLASTIC LABEL 1-11 SLOT LABEL	1
9	653-411-01	PLASTIC LABEL 0-10 SLOT LABEL	1
10	656-411-00	PLASTIC, UF2 TH SBR LABEL, LABEL	1
11	638-517-04	PLASTIC CDU COOLANT RETURN LABEL	1
12	638-517-03	PLASTIC CDU COOLANT SUPPLY LABEL	1
13	639-933-00	ALUMINUM 10G CABLE HOLDER BRACKET OR MISC ARTIC	4

METHOD:

Flow chart of operation steps

BPM preparation

Preparation Tools

Prepare Materials

Complete Operation

Inventory Materials

Operation Check

Inventory tool

TOOL NAME	TORQUE VALUE
PH2 Phillips torque screwdriver	3.3 N·m
17mm socket wrench	3.3 N·m
T20 Torx wrench	3.3 N·m
4mm hexagon socket head torque	3.3 N·m
Hose cutter	NA

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION				QTY
1	638-907-01	ELEC-MECH IPD VALVE SHUTL VER FOR SEMI TEST				1
2	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W/CONE				2

MACHINE:

PH2 Phillips torque screwdriver 3.3 N·m.

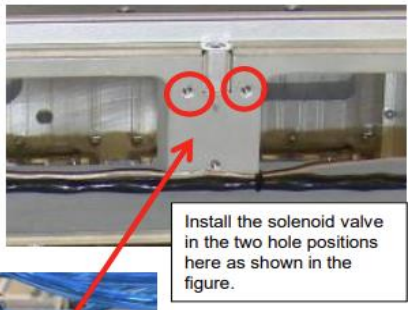
METHOD:

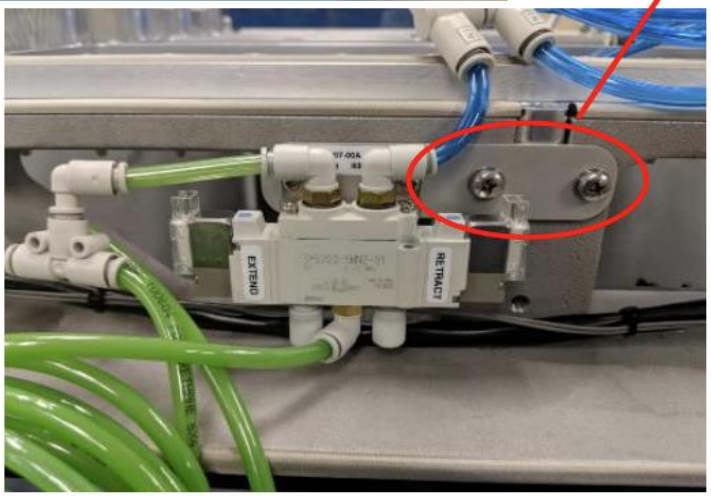
Operation steps

1.1 Take one IPD solenoid valve (638-907-01).

1.2 Rotate the T/H direction as shown, and install an IPD solenoid valve (638-907-01) on the T/H frame, Fix the IPD solenoid valve with 2 M5 screws (472-426-12). The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m

Note: During installation, make sure that the black wire is pushed behind the IPD solenoid valve and not pressed.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

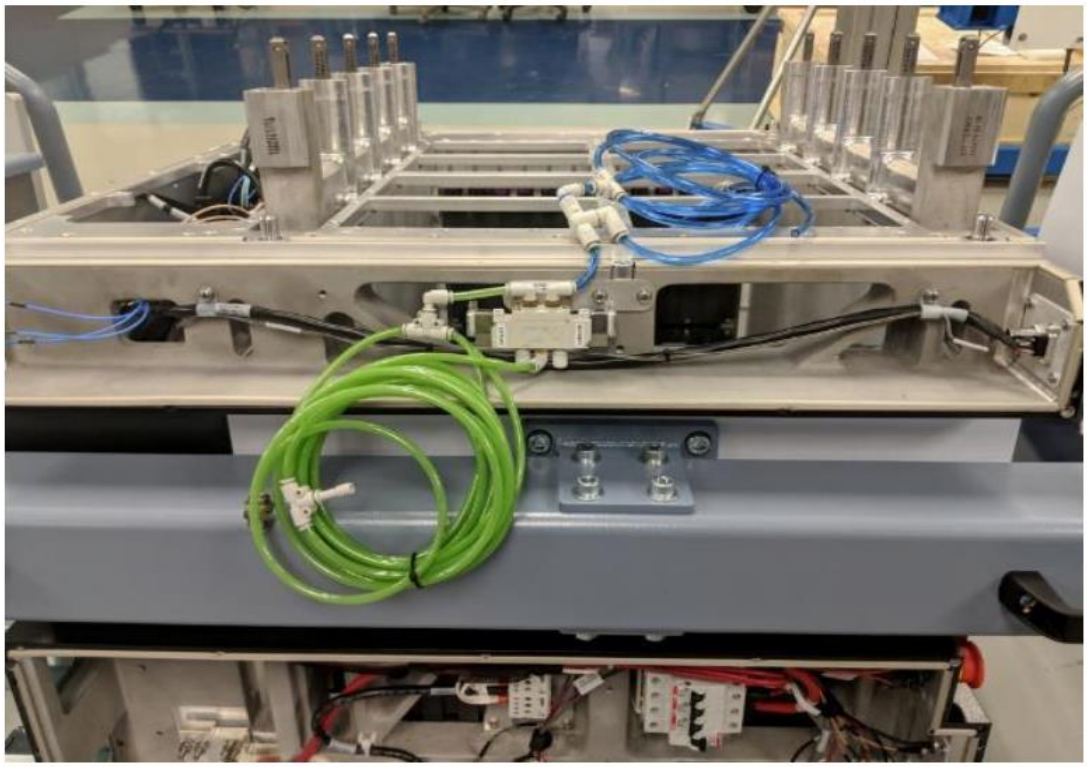
N/A

MACHINE:

N/A

METHOD:

1.3 Make sure that the hoses and accessories are as shown in the figure below. On the right is the blue hose, on the left is the green hose.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

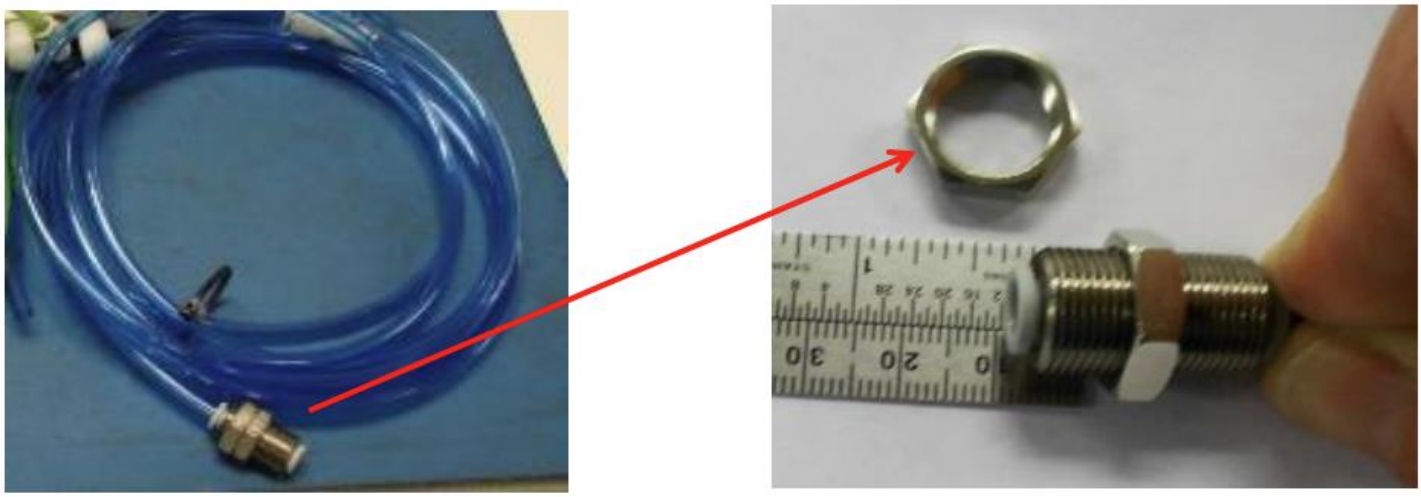
N/A

MACHINE:

N/A

METHOD:

1.4 Remove the front nut of the blue external trachea connector as shown in the figure, and adjust the distance of the nut behind the trachea connector to a distance of 10 mm from the tube connector.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

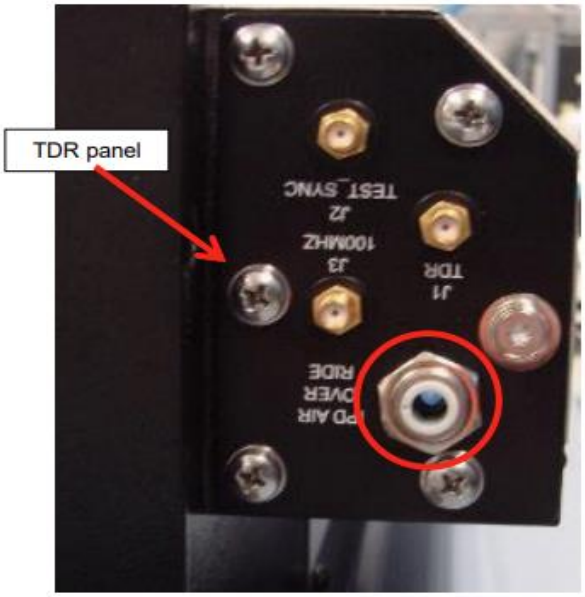
17mm socket wrench 3.3 N·m, PH2 Phillips torque screwdriver 3.3 N·m

METHOD:

1.5 Insert the trachea connector into the TDR panel as shown in the figure, and install the nut, and fix the nut with a 17mm socket wrench, the torque value is 3.3 N·m.

If the installation nut interferes, the lower and upper screws on the panel can be removed first. After installing the nut put back the screw.

Use a PH2 Phillips torque screwdriver to fix the screws again. The torque value is 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

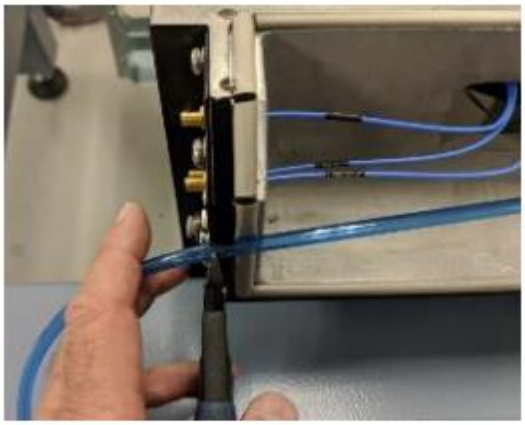
N/A

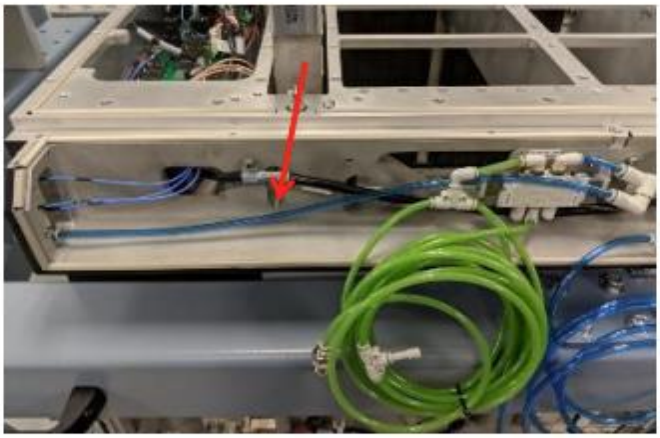
MACHINE:

N/A

METHOD:

1.6 Orient the end of the hose toward the TDR panel as shown in the figure, cut off the excess part with a suitable length, and insert the hose into the connector just installed





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
3	639-820-00	CABLE IPD CONTROL W/ CONN	1

MACHINE:

N/A

METHOD:

2.1 Take the IPD control cable (639-820-00) as shown in the figure, pass one end of the large cable connector through the T/H opening slot and place it in the backplane area.
(This connector will connect the PTL LCL board in subsequent assembly.)

2.2 Insert the end of the line label "Retracted/Extended Cylinder Rear" into the T/H square slot on the LEFT side of the solenoid valve as shown in the figure to enter the T/H frame.

2.3 Insert the end of the line label "Retracted/Extended Cylinder Front" into the T/H square slot on the RIGHT side of the solenoid valve as shown in the figure, and enter the T/H frame.

2.4 Use the existing cable ties to organize and tie the IPD wires and tubes.

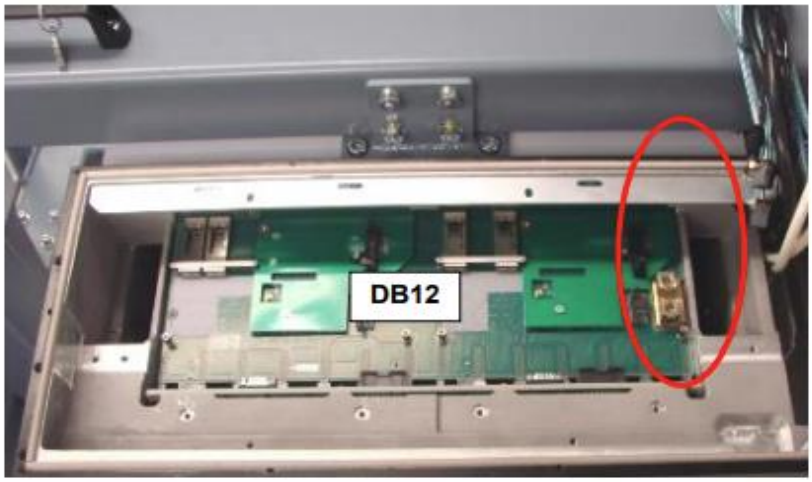
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

j. The wiring method is as shown in the figure, reserve enough long wires to insert into the MP12 and DB12 PCBA boards.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	813-929-01	HDW VELCRO STRP 5/8	2

MACHINE:

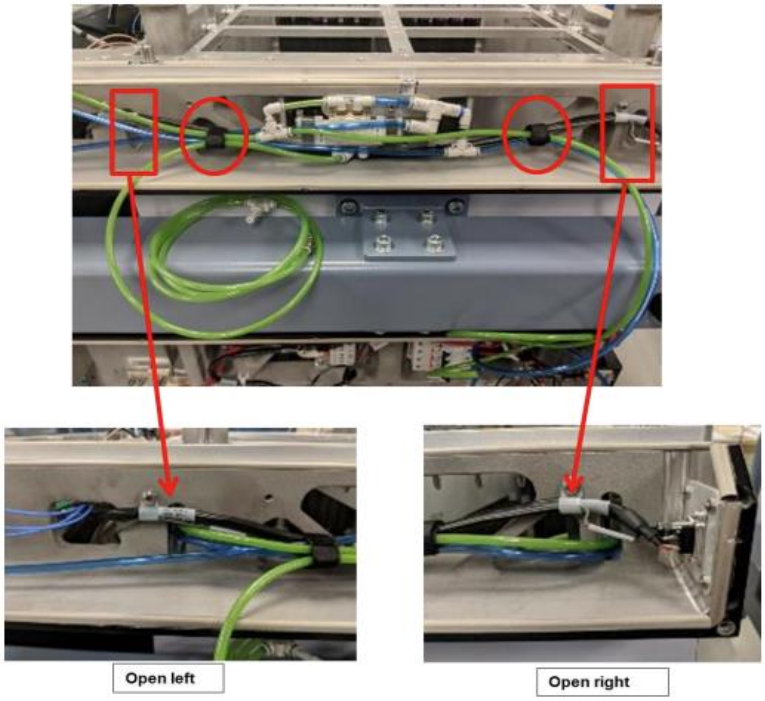
N/A

METHOD:

1.7 Insert a set of trachea (blue, green) into the opening slot on the left side of the solenoid valve as shown, and enter the T/H frame.
and enter the T/H frame.

1.8 Insert another set of trachea (blue, green) into the open slot on the right side of the solenoid valve as shown,
and enter the T/H frame.

1.9 Use two cable ties (813-929-01) to tie the two trachea together with the existing wire harness
as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

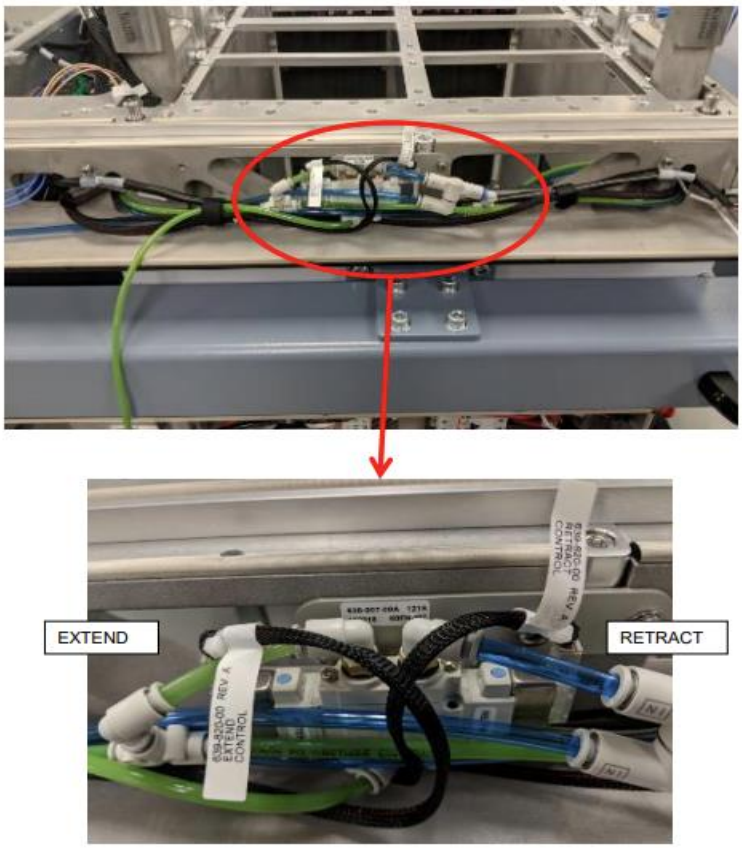
N/A

MACHINE:

N/A

METHOD:

2.5 Insert the connector of the line label "Extend/Retract" into the solenoid valve "Extend/Retract" as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

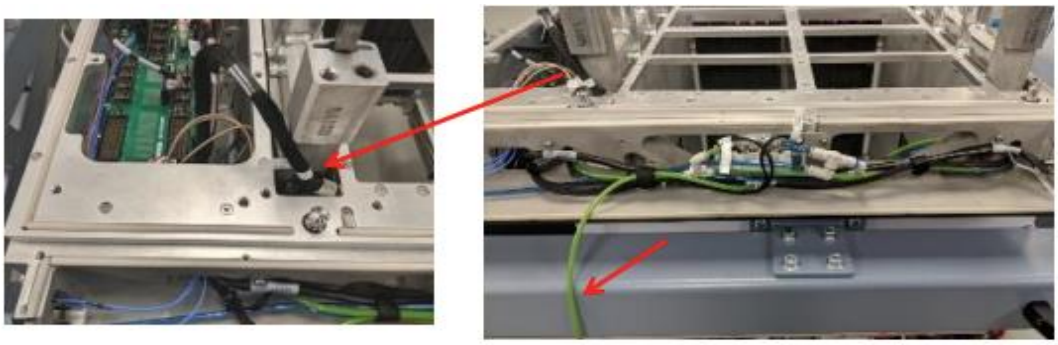
N/A

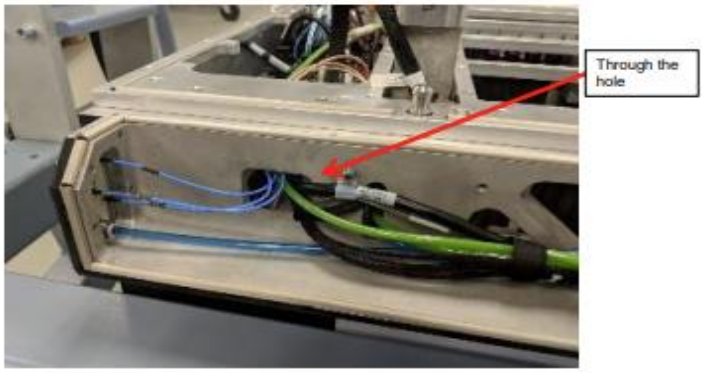
MACHINE:

N/A

METHOD:

2.6 As shown in the figure, pass the green tube through the small hole to the backplane area, temporarily hang the wire first, and then pass it to the disconnection panel area.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

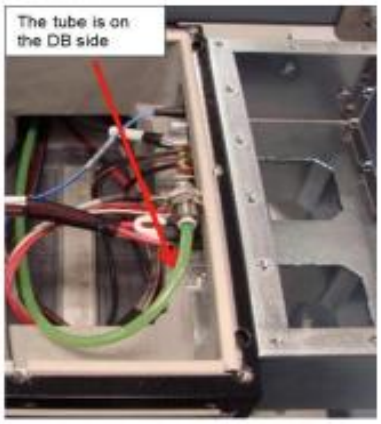
N/A

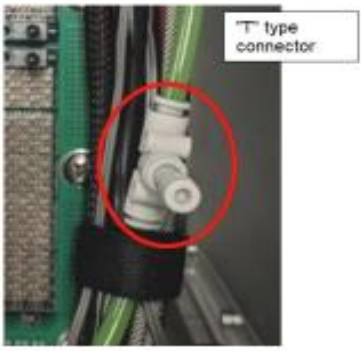
MACHINE:

N/A

METHOD:

2.7 Place the green trachea through the Manifold block, penetrate the channel in the T/H frame and place it on the DB surface as shown in the igure.
Use the existing cable tie to tie the trachea and the harness together, and place the "T" fitting upwards in the position shown on the right.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	634-338-00	ALUMINUM COVER AIR HOSE BRACKET OR MISC ARTICLE	2

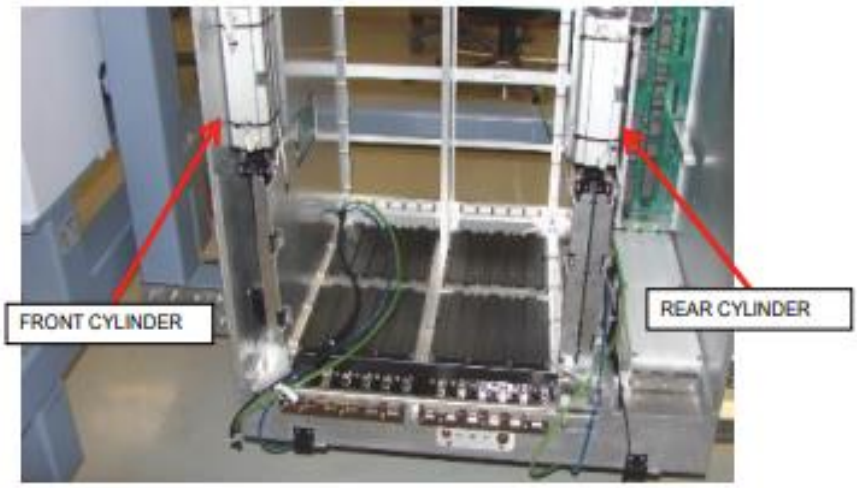
MACHINE:

N/A

METHOD:

3.1 As shown in the figure, the positions of the IPD Retracted/Extender Cylinder Rear and Retracted/Extender Cylinder Front threads are within the frame.

3.2 Take 2 IPD brackets (634-338-00), put the blue trachea, green trachea and control wire into the bracket.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

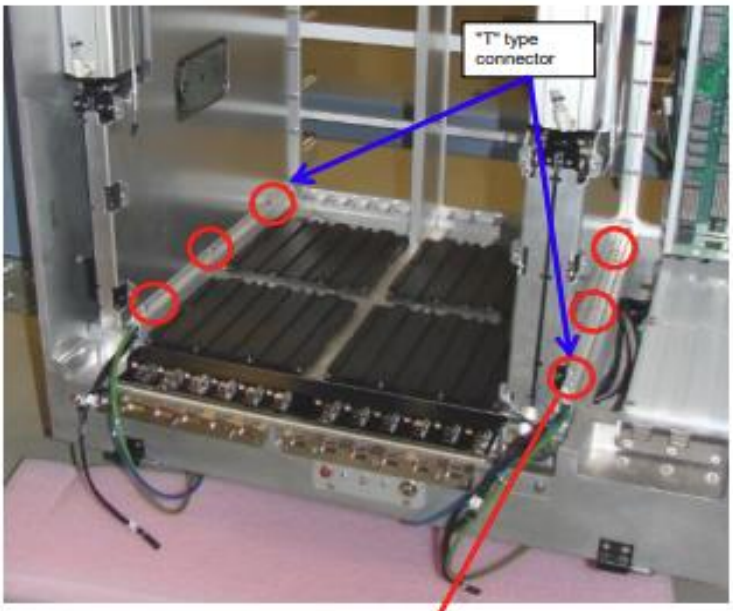
ITEM	PART NUMBER	DESCRIPTION	QTY
6	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	6

MACHINE:

T20 Torx wrench 3.3 N·m

METHOD:

3.3 Use 6 M5 screws (406-502-12) to fix the bracket on the T/H frame. The tool is a T20 Torx wrench and the torque value is 3.3 N·m.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
7	813-929-01	HDW VELCRO STRP 5/8	3
8	813-929-00	**HDW VELCRO STRP 5/8**	1

MACHINE:

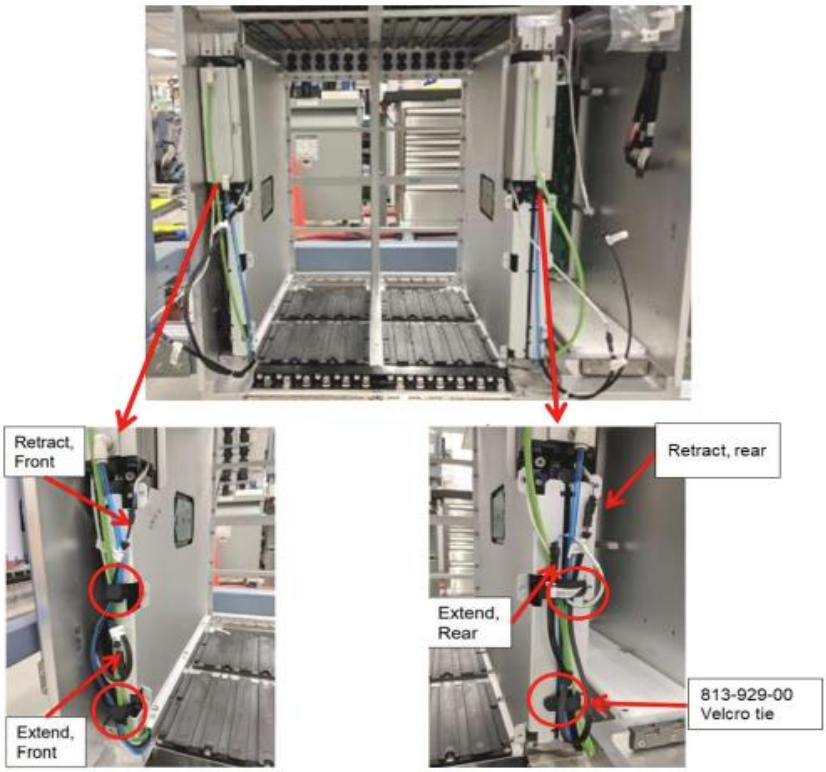
N/A

METHOD:

4.2 Insert the blue and green gas pipes into the cylinder as shown in the picture. The top is green and the bottom is blue. If it is too long, cut off the excess.
Adjust the elbow joint on the cylinder to the vertical direction.

4.3 Connect the white line of the cylinder according to the label of the IPD control line as shown in the figure, a total of 4 positions.

4.4 As shown in the figure, use three 813-929-01 cable ties and one 813-929-00 cable ties to tidy up the signal cables and trachea.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
9	813-929-01	HDW VELCRO STRP 5/8	2

MACHINE:

N/A

METHOD:

4.5 Use another cable tie (813-929-01) at the position shown in the figure to tie the trachea and the control wire together, keeping the cable and the trachea away from the metal edge.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

4mm hexagon socket head torque wrench 3.3 N·m.

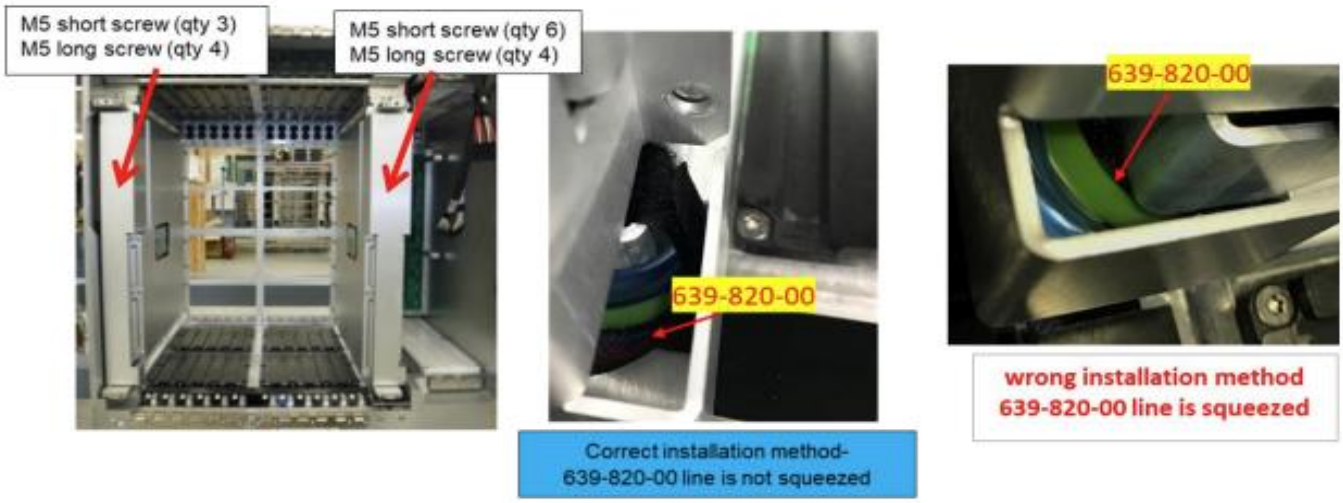
METHOD:

4.6 One (657-144-00) IPD cover plate and one (650-469-00) IPD cover plate removed before installation as shown in the figure, and four expansion blocks.

Use the remaining 9 M5 short screws to fix the cover plate and 8 M5 long screws to fix the expansion block.

The tool is a 4mm hexagon socket head torque wrench with a torque value of 3.3 N·m.

Note: Make sure that the trachea and IPD control cable are all inside and not caught by the cover when installing the cover!



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

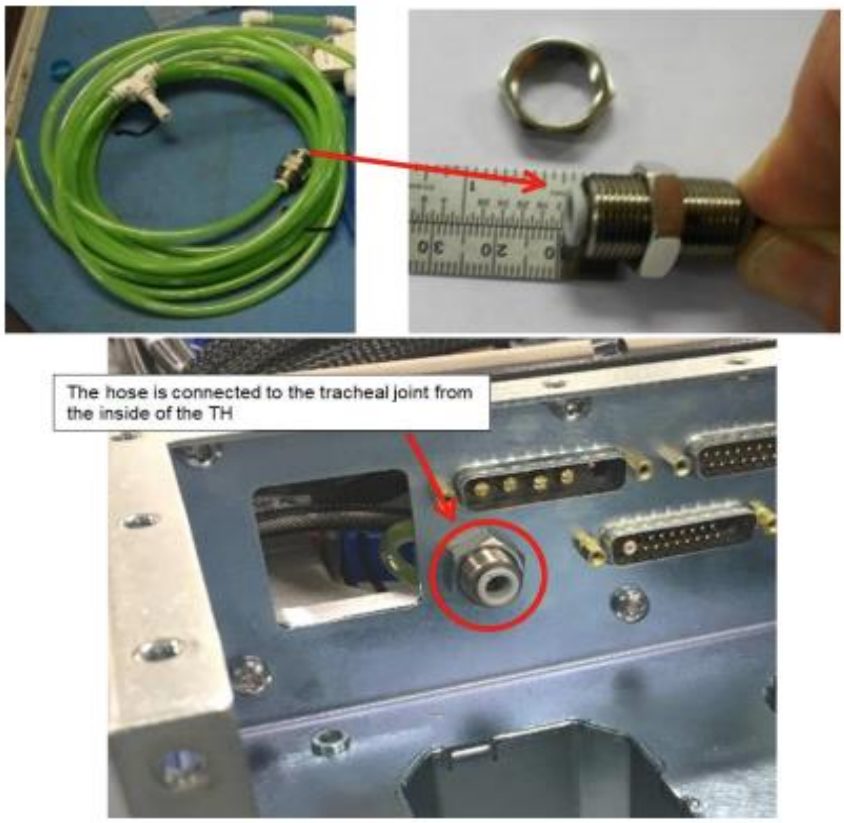
17mm socket wrench 3.3 N·m

METHOD:

5.1 Remove the nut on the front of the green trachea as shown in the figure, and then adjust the nut on the back, 10mm away from the front end.

5.2 Install the trachea connector to the panel as shown in the figure, and then install the nut to fix it. The tool is a 17mm socket wrench and the torque value is 3.3 N·m.

5.3 Connect the trachea from the inside of TH to the trachea connector as shown in the figure.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

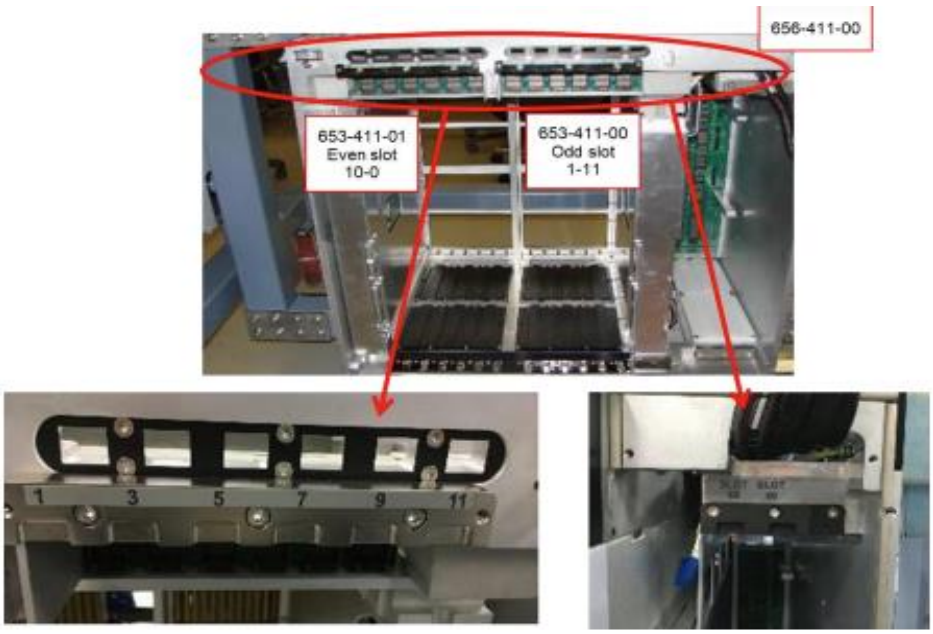
ITEM	PART NUMBER	DESCRIPTION	QTY
10	653-411-00	PLASTIC LABEL 1-11 SLOT LABEL	1
11	653-411-01	PLASTIC LABEL 0-10 SLOT LABEL	1
12	656-411-00	PLASTIC, UF2 TH SBR LABEL, LABEL	1

MACHINE:

N/A

METHOD:

6.1 Take the following labels:
a. 1 UF2 TH SLOT 1-11 label (653-411-00).
b. 1 UF2 TH SLOT 0-10 label (653-411-01).
c. 1 UF2 TH SLOT 66,48 label (656-411-00).
6.2 Rotate the T/H direction as shown. Note that the red circle is the location of the label.
6.3 Install the label on the T/H frame as shown.
6.4 Install the label position as shown in the figure, the label paper is in the center, and the top position of the label is kept flush with the metal edge. Repeat the installation of the label at several other locations.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
13	638-517-04	PLASTIC CDU COOLANT RETURN LABEL	1
14	638-517-03	PLASTIC CDU COOLANT SUPPLY LABEL	1

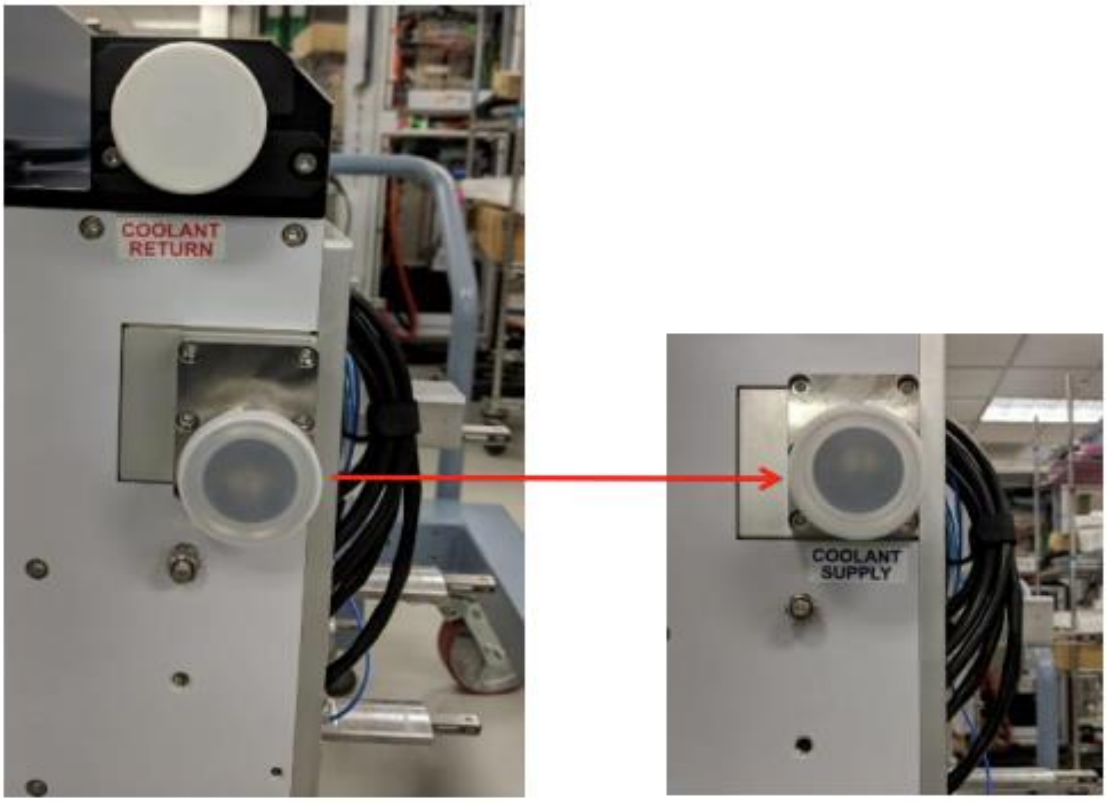
MACHINE:

N/A

METHOD:

7.1 Take one Coolant Return label (638-517-04) and paste it to the position shown in the figure.

7.2 Take one Coolant Supply label (638-517-03) and paste it to the position shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

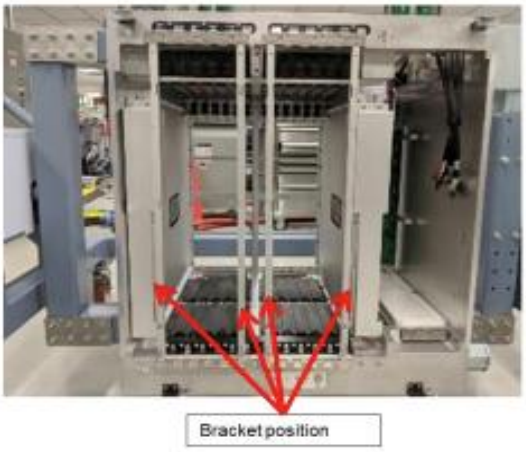
ITEM	PART NUMBER	DESCRIPTION	QTY
14	639-933-00	ALUMINUM 10G CABLE HOLDER BRACKET OR MISC ARTICLE	4

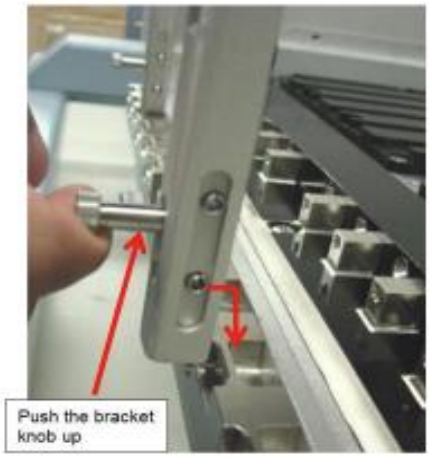
MACHINE:

N/A

METHOD:

7.1 Take 4 wire support brackets (639-933-00).
7.2 Install the wire support brackets on the maintenance surface 0, 10, 1, 11 slot position (slot 0 and slot 10 are even numbers, slot 1 and slot 11 are odd numbers).
7.3 Install the top of the cable bracket into the latch lever cavity as shown in the figure.
7.4 Install the bottom of the cable bracket into the latch lever cavity by pushing up the bracket knob.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Matters needing attention

1. Do ESD and protective safety measures.
2. Be careful during the assembly process so as not to scratch the surface of the parts.
3. After assembly, check the assembled parts for defects and leak-free operation.